

APPLE INC.: FIGHTING PATENT TROLLING¹

Arpita Agnihotri and Saurabh Bhattacharya wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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Version: 2020-09-18

By February 2020, Apple Inc. (Apple) was fighting litigation from several patent troll² firms that had accused Apple of patent infringement involving several technologies.³ One of Apple's oldest litigation battles was with VirnetX Holding Corporation (VirnetX), a US-based software developer and technology solution provider, which had first sued Apple for patent infringement in 2010, in a patent troll-friendly court in eastern Texas.⁴ In February 2020, a US federal court had to decide whether Apple would be required to pay VirnetX for patent infringement.⁵ Apple, along with other technology companies such as Samsung and Intel Corporation (Intel), appealed to the US government and to the European Union to take action against patent trolls.⁶ Patent trolls were making it difficult for companies to innovate due to the risk of potential litigation.⁷

Tim D. Cook, chief executive officer (CEO) of Apple said, "The US court system is currently structured in such a way that it's difficult to get the protections a technology company needs because the cycle is very long. For us, our intellectual property is so important to our company. I would love to see the system strengthened to protect it."⁸ Apple itself was fighting several litigations against patent troll firms such as VirnetX, Straight Path IP Group Inc., and Uniloc USA, Inc.,⁹ and was also suing them for the potential damage they were doing to the company's economic and innovative abilities.¹⁰ Patent troll firms, on the contrary, believed that they were promoting innovation by protecting the rights of individual inventors.¹¹ Companies such as Hewlett-Packard Company (HP) and Oracle Corporation (Oracle) which had "giant stacks of patents," claimed they were ready to use these "to crush the competition if they deem it necessary."¹² Apple asserted that trolling was adversely impacting their economic and innovation performance.¹³ How could Cook ensure that the company had the proper systems in place to stop trolling? How should he fight the battle against patent trolling? How could he minimize the adverse effects of trolling on the innovative and economic performance of Apple?

BACKGROUND

Apple was one of the world's most valuable companies¹⁴ and consequently was the number-one target for attempted lawsuits from non-practicing entity (NPE) firms, or patent trolls.¹⁵ In 2019, Apple had revenue of US\$260 billion¹⁶ and 137,000 employees.¹⁷ In 2012, it had 1,435 patents in force globally, while in 2017, the company had 473 in-force patents¹⁸ (see Exhibit 1). As one researcher pointed out, "a whopping 9 out of 10 patent lawsuits filed against tech companies in the first half of 2015 were brought by NPEs."¹⁹

VirnetX's business model rested "solely on suing companies that actually develop and sell products," and the company took advantage of "arcane patent infringement laws and loose interpretations of intellectual property regulations."²⁰ In a US Securities and Exchange Commission filing, VirnetX said its business model was founded on a "portfolio of intellectual property."²¹ In early 2020, VirnetX's patent portfolio included over 80 US and international patents and over 100 pending applications. Over the years, VirnetX had stated its intention to market various products, including a "secure domain name registry," but its revenues came primarily from licensing patents.²² In early 2020, VirnetX had 20 employees²³ and a small office suite, the rent for which was \$5,000 a month.²⁴

Apple asserted "that NPEs make no products, employ few or no people, create no jobs, and exploit the US patent system to seek royalties for technology invented, most often, by Apple."²⁵ In 2019, most patent litigations against corporations were filed by patent trolling firms, especially in the high-tech sector (see Exhibits 2 and 3). Seeing the potential in patent trolling, lawyers themselves were forming trolling companies. For example, Blackbird Technologies Inc., a Boston- and Chicago-based firm, was formed by two attorneys who "amassed dozens of patents, then began using them to file dozens of patent infringement lawsuits."²⁶ In 2016, IP Edge, an intellectual property monetization firm in Texas, was the most active patent troll firm, pursuing 417 litigations.²⁷ In 2018, SoftBank Inc., an investment company, began suing tech companies through its Fortress Investment Group arm for intellectual property infringement.²⁸

EASTERN TEXAS: THE HUB OF PATENT TROLL FIRMS' LITIGATION

Until 2017, the Eastern District of Texas was the site of thousands of patent case filings each year (see Exhibit 4). In 2015, this number stood at 2,500, and 95 per cent of these suits were filed by NPEs.²⁹ Most patent cases in the Eastern District of Texas were filed in the town of Marshall, which had a population of 24,000. By 2017, 25 per cent of all patent litigations in the United States were filed in Marshall; patent litigation had become the engine that drove Marshall's economy.³⁰ Nathaniel Polish, of Daedalus Technology Inc., said, "At any given moment, there are six to ten teams of lawyers staying in Marshall."³¹ Because hotels in Marshall were always full but had no restaurants, locals started cooking and offering catering services. Some served as couriers as well. Marshall also experienced a construction boom, as more office space was created for the legal teams, who, Polish added, "bring in a lot of money."³² The American Tort Reform Association labelled the Eastern District of Texas one of its "judicial hellholes"³³ and complained that "the court is viewed as favourable to so-called 'patent trolls' by refusing to dismiss meritless claims early in litigation, thereby providing claimants with greater settlement leverage."³⁴

However, in 2017, the US Supreme Court "cracked down on the practice"³⁵ by requiring firms claiming patent infringement to file the case where the charged firm was headquartered or conducted its regular business,³⁶ rather than "hunting for a friendly court."³⁷ This was considered "bad news for patent trolls, . . . which make no products but brandish patents to sue companies for infringement" rather than conducting any actual research or development.³⁸ By 2019, although Delaware and the Central District of California had become the number-one and -two litigation destinations in the United States, the Eastern District of Texas remained one of the major hubs for patent litigation³⁹ (see Exhibit 4).

DRIVERS OF PATENT TROLLS

Patents were generally granted for 20 years, beginning from the time a company applied for the patent, at a cost of \$10,000. James D. Crowne, the deputy executive director for legal affairs of the American Intellectual Property Law Association stated that "the whole idea of exclusive rights in the patent law is to provide an incentive for innovation to arise, and it's a limited right."⁴⁰ However, patent fees could exceed

\$25,000, and “at least a couple years of effort” were required to get a patent. Sometimes, patents granted by the US Patent and Trademark Office (PTO) were subsequently litigated, and courts declared them invalid, implying the patents should not have been granted at all.⁴¹ This happened if an invention in a given scientific or technical field was deemed to be obvious to anyone familiar in the field.⁴² According to experts, such patents were granted because PTO officers did not have ample time to review the applications. In 2018, the PTO received 643,349 patent applications, or 1,763 applications per day. The agency’s roughly 8,185 patent examiners were overburdened.⁴³

Trolls took advantage of vague, difficult-to-interpret words and phrases in patents to make claims that otherwise were difficult to predict.⁴⁴ Thus, even if the accused infringer read the patent closely, it was difficult to understand how an infringement had happened. Furthermore, a plaintiff filing a lawsuit was not required to be specific about how infringement had taken place or exactly which patents had been copied.⁴⁵ Neither were weak infringement claims scrutinized by the courts to immediately turn away frivolous claims. As statements in the legal complaints were vague, defendants found it difficult to quickly or easily explore all potential defences and responses.⁴⁶ Furthermore, patent trolls specifically targeted only cash-rich firms. If a firm was diversified and cash rich, patent troll firms sued them for patent infringement even if the division concerned was losing money.⁴⁷ In early 2020, Apple was one of the most cash-rich companies in the world (see Exhibit 5).

NON-PRACTICING ENTITY (NPE) LAWSUITS ALLEGING PATENT INFRINGEMENTS BY APPLE

In February 2020, Apple was sued by Neodron Limited (Neodron), an Irish patent-holding company, for two patents involving functions of capacitive keyboards. Neodron stated that Apple had used its keyboard technology, patented in 2010 and 2011, in the design of the iPhone 11.⁴⁸ Neodron, which was formed in late 2018, sued several leading tech companies—including Amazon.com Inc. (Amazon), Dell Computer Corporation (Dell), HP, and Samsung—for patent infringement,⁴⁹ alleging that touchscreen devices from these companies used technology that infringed its patents. Neodron had not invented or created new products or technological advances before accusing these companies of patent infringement but had only acquired a few patents.⁵⁰

Several international companies also opened offices in Texas to sue Apple for patent infringement. Seven Networks Inc., of Finland, set up a US headquarters in Texas, transferred its patents there, and then sued Apple for infringement of over 16 patents in April 2019.⁵¹ The alleged infringements covered a range of issues, from the way iCloud operated to the features of specific devices.⁵²

Since January 2020, Apple had also been fighting litigation from Straight Path IP Group Inc. in Europe “over four patents that relate to video conferencing and point-to-point communications between devices.”⁵³ Until 2019, two firms connected with Fortress Investment Group, Uniloc USA and Uniloc Luxembourg, filed at least 25 lawsuits against Apple and demanded \$2.6 billion to \$5.1 billion in damages. Fortress Investment Group argued that it was entitled to as much as \$2.75 for every iPhone sold.⁵⁴ The two Uniloc firms were, in fact, some of the most active NPEs in the world, and were eventually featured in a documentary, *The Patent Scam*, available on Amazon Prime Video.⁵⁵ In February 2020, NavBlazer, another patent troll company, sued Apple for using a navigation system in its iPhones that purportedly infringed on patents acquired by NavBlazer.⁵⁶

Among the several patent troll litigations Apple was fighting, the one with VirnetX was the oldest. VirnetX, based in Zephyr Cove, Nevada, filed lawsuits against Apple in the patent troll-friendly Eastern District of Texas, where patent plaintiffs had a higher chance of success in intellectual property cases.⁵⁷ VirnetX

charged Apple with patent infringement relating to secure communications technologies, including FaceTime, which Apple used in its products.⁵⁸ Although VirnetX demanded \$900 million in penalties, the jury decided in 2012 on a figure of \$368 million.⁵⁹ In 2016, VirnetX again filed a lawsuit against Apple, alleging that Apple's VPN technology and FaceTime video messaging system infringed VirnetX patents and claiming \$532 million in damages.⁶⁰ As the patent war continued, in April 2018, a federal court in Texas ordered Apple to pay a penalty of \$502.6 million to VirnetX for violating "four patents related to Internet-based communications" that VirnetX claimed it owned; Apple asked a US court of appeals to reconsider this decision.⁶¹ In November 2019, "a three-judge panel voided that \$502.6 million calculation, granting Apple a win of sorts" against VirnetX. However, judges maintained the previous jury's finding that some older iPhone models did infringe two VirnetX patents.⁶² The Texas court had to recalculate how much Apple would need to pay VirnetX, and there was a possibility that the court could call for a complete retrial, as Apple's patent infringements were reduced from four to two.⁶³

IMPACT OF PATENT TROLLS

A survey conducted by the American Intellectual Property Law Association in 2019 reported that, between 2009 and 2019, the "cost of patent litigation in US district courts . . . largely remained constant."⁶⁴ According to Scott McBride, a partner at McAndrews, Held & Malloy Ltd., the survey indicated that "in 2009, the median patent litigation cost through post-trial (fees, expenses, and costs) was \$650,000 to \$5.5 million. By 2013, that range was \$700,000 to \$5.5 million. In 2019, the range was \$700,000 to \$4 million."⁶⁵

Patent litigation was common when corporations felt that competitors had infringed their patents.⁶⁶ Steve Jobs, in fact, had accused Google of stealing Android technology from Apple and said he was "willing to go thermonuclear war on this."⁶⁷ Apple, in a complaint to the US District Court for the Northern District of California in 2019, mentioned that it had "suffered economic harm in the form of litigation costs and diversion of resources away from innovation to respond to these entities."⁶⁸ A Harvard Business School study conducted in 2016 revealed that patent troll settlements had resulted in a 25 per cent decrease in research and development (R&D) by large corporations.⁶⁹ Another study, conducted by Catherine Tucker of the Massachusetts Institute of Technology in 2011, found that medical imaging software firms involved in patent troll litigations had reported a one-third decline in sales compared to those not so involved. This happened as firms suffering from patent troll litigations stopped any incremental innovation on devices involved in litigation.⁷⁰

Patent Strategy of Firms

For some firms, the motivation for patenting changed over the years. Rather than applying for patents in order to prevent imitation by competitors, companies were filing patents for minor innovations to create portfolios of patents they could then use in cross-licensing arrangements with other players in the industry. Although this required significant effort, time, and money, the tactic not only hindered new entrants in the field but also encouraged cooperation among competitors.⁷¹ Another competitive tactic was filing for patents not to prevent imitation but to ensure that competitors were not able to innovate. Companies filed such patents in areas that were not even their focus, and this reduced economic efficiency because such companies neither offered products and services based on the filed patents nor let their competitors innovate in these areas.⁷²

In fact, many large corporations had themselves indulged in buying patents to enhance their portfolios instead of innovating. Since 1920, the International Business Machines Corporation (IBM) had acquired more than 140,000 US patents, and in 2016 and 2019, claimed more than 8,000 and 9,000 patents, respectively. IBM was patenting large quantities of "fluff," rather than engaging in real innovation.⁷³ In

2017, IBM even had a patent for the “out-of-office” email system.⁷⁴ Critics at this point considered IBM to be a patent troll firm.⁷⁵ According to critics, IBM spent \$6 billion on R&D in 2019,⁷⁶ while Apple spent \$13 billion in the same year, making it one of the largest R&D spenders globally.⁷⁷ However, among major tech firms, Apple received one of the lowest numbers of patents in the United States in 2019, while IBM received the most (see Exhibit 6). HP and Oracle also had numerous patents, and according to media reports, they were ready to use them to damage their competitors if necessary.⁷⁸

Smaller firms, on the other hand, suffered because of patent trolls as venture capitalists stopped investing in innovative firms beyond a certain number of litigation cases.⁷⁹ However, the patent troll firms said they were promoting innovation only among small inventors: NPEs claimed that, after they purchased patents from small inventors who did not have resources to fight large firms such as Apple, they could share a portion of the payments obtained by winning lawsuits against the big corporations with these inventors.⁸⁰

Given these inefficiencies, experts demanded the complete abolition of the patent system and its replacement with some other system like open innovation, which was less exploitative.⁸¹ Patents decreased social welfare and promoted monopolies, but with patent trolls, this loss of social welfare was further worsened. Though there was a risk of demotivation with this approach, experts believed that, if government offered grants or incentives for open innovation, the problems of patent trolls could be resolved.⁸²

APPLE'S INITIATIVES AGAINST PATENT TROLLS

After the Supreme Court decided in 2017 that a defendant could be sued only in a district where it “resided”—where it was incorporated or had “a regular and established place of business”—Apple decided to close its retail stores in the Eastern District of Texas.⁸³ This helped Apple better protect itself from patent infringement lawsuits. This move was considered beneficial despite the fact that Apple had invested almost \$30 million in its Dallas-area stores.⁸⁴ Apple offered retail employees affected by the changes new jobs with the company.⁸⁵

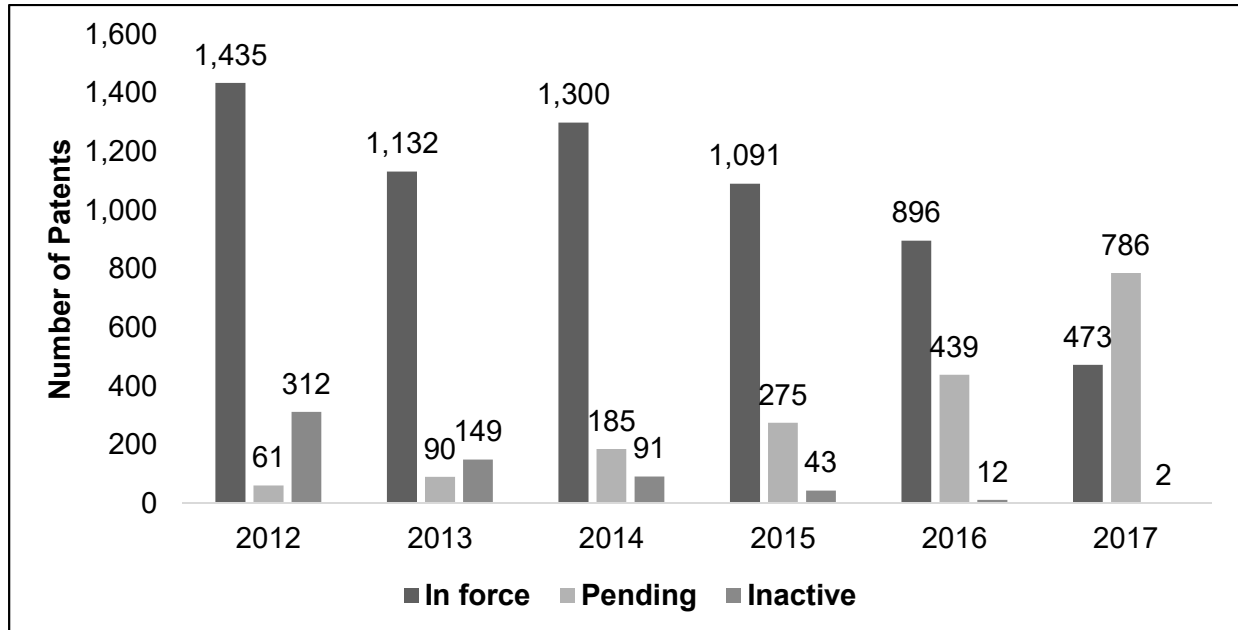
Apple also took several initiatives to fight directly against the patent troll companies, rather than merely avoiding litigations. In 2018, Apple cooperated with Samsung, even though the two companies were fighting a patent war against each other, to convince a US federal court to invalidate a patent related to mobile hotspots created by smartphones.⁸⁶ When IXI Mobile Inc., another patent troll company, claimed it was owed licensing fees from Apple, Samsung, and other phone makers, the two companies were jointly able to convince a US federal court that this patent should not have been granted in the first place, as it was not an original idea.⁸⁷ IXI Mobile Inc. claimed it held a patent on the entire concept of a mobile hotspot: its patent number 7,039,033 covered the “system, device and computer readable medium for providing a managed wireless network using short-range radio signals.”⁸⁸

Apple also joined forces with Intel to stop patent trolls. In November 2019, Apple and Intel filed an antitrust lawsuit against Fortress Investment Group, charging that the group’s existence was based entirely on patent trolling, as it had no shipping product or service to offer as part of its business.⁸⁹

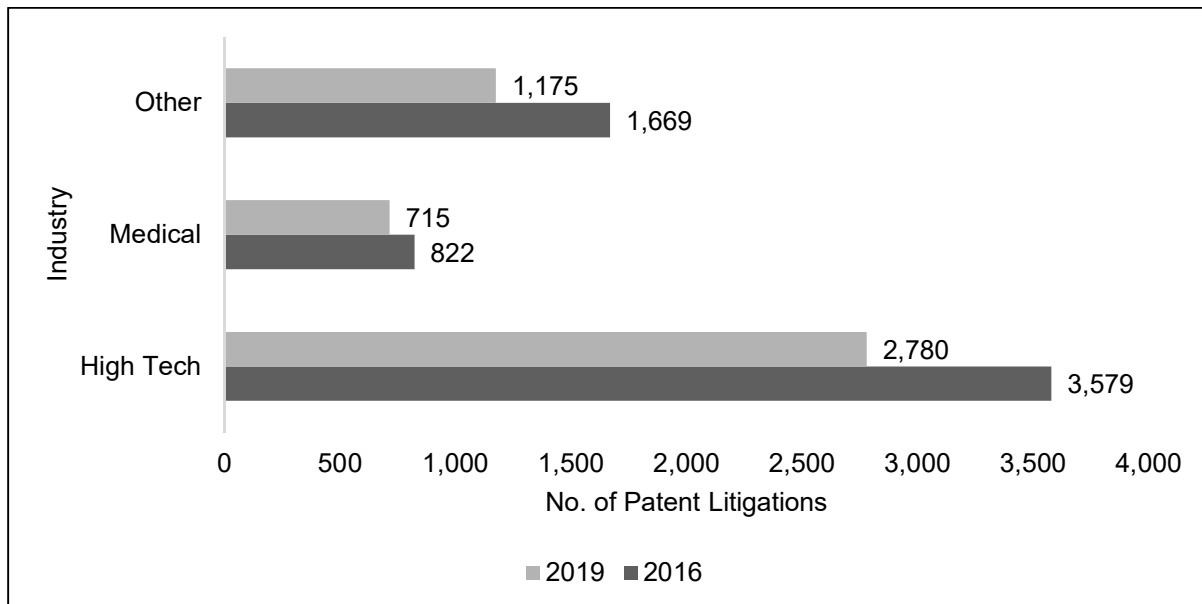
In early February 2020, Apple filed a pre-emptive “Complaint for Declaratory Judgment of Non-Infringement” against Fundamental Innovation Systems International (FIS). This kind of lawsuit was filed when a defendant was threatened with a patent infringement lawsuit. A declaratory judgment permitted the defendant to seek to have the patent declared invalid or not infringed.⁹⁰ Apple was aware that FIS had already sued LG Electronics Inc., Huawei Technologies Co., Ltd., and Zhongxing Telecommunication Equipment Corporation and could sense another impending lawsuit. Apple referred to 12 patents by BlackBerry and Research in Motion, which Fundamental Innovation Systems accused were infringing and declared that “Apple’s products do not infringe the Patents-in-Suit.”⁹¹

THE ROAD AHEAD

When faced with frivolous lawsuits from patent trolls, several companies preferred to settle rather than embark on costly litigations.⁹² Apple, however, took a more principled stand, fighting back against NPEs. By January 2020, Cook had urged government authorities in the European Union to act against patent trolls.⁹³ Apple joined a coalition of 35 companies in Europe and four industry groups, which included Microsoft and BMW, to write to Thierry Breton, the European Union Commissioner for Technology and Industrial Policy, about patent trolls, which were abusing existing regulations.⁹⁴ In February 2020, the Apple–VirnetX litigation rested with a Texas court. VirnetX was expected to earn millions if it won. However, the business model of patent troll firms was not expected to remain viable in the long term.⁹⁵ The US Congress had introduced several bills to regulate the licensing and assertion of patents and was considering the Innovation Act (H.R. 9 and H.R. 3309) and the Targeting Rogue and Opaque Letters Act (H.R. 2045), among others, while the USPTO [United States Patent and Trademark Office] undertook an initiative to enhance patent quality.⁹⁶ How should Cook dethrone the patent troll firms? How could he ensure constant innovation in the uncertain climate of patent trolling?

EXHIBIT 1: NUMBER OF APPLE PATENTS WORLDWIDE (2012–2017)

Source: Nils-Gerrit Wunsch, "Number of Apple Patent Families Worldwide by Filing Year from 1998 to 2018, by Legal Status," Statista, July 2019 updated September 8, 2020, accessed March 25, 2020, www.statista.com/statistics/1033935/number-of-apple-patents-by-filing-year-and-status-worldwide/.

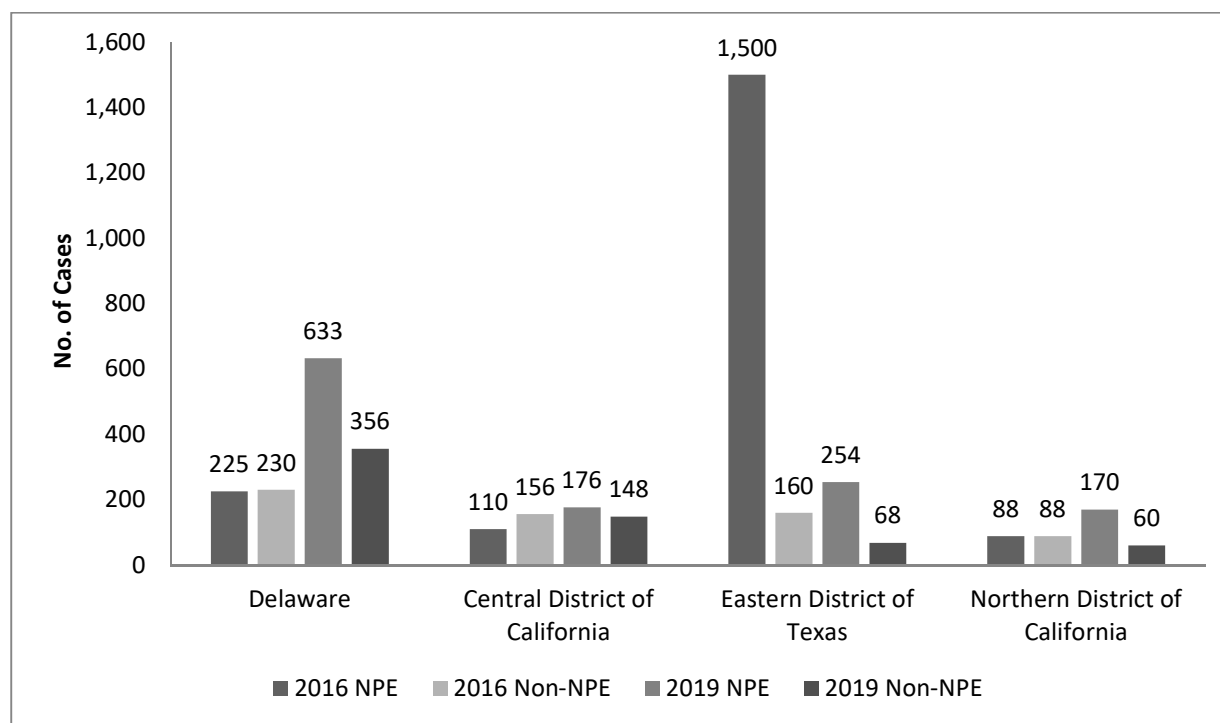
EXHIBIT 2: NUMBER OF PATENT LITIGATIONS, BY INDUSTRY (2016, 2019)

Source: Created by case author based on data from "2019 Patent Dispute Report—Year in Review," UnifiedPatents, January 1, 2020, accessed March 25, 2020, www.unifiedpatents.com/insights/2019/12/30/q4-2019-patent-dispute-report; "2016 Annual Patent Dispute Report," UnifiedPatents, January 1, 2017, accessed March 25, 2020, www.unifiedpatents.com/insights/2016/12/28/2016-annual-patent-dispute-report.

EXHIBIT 3: PERCENTAGE OF DISTRICT COURT LITIGATIONS, BY INDUSTRY (2016, 2019)

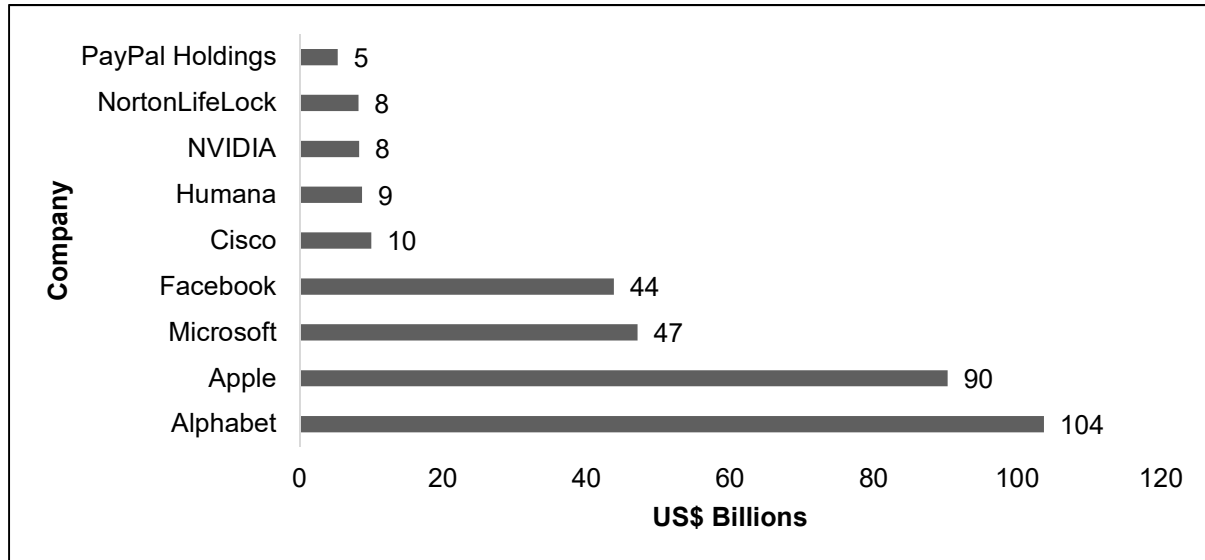
Industry	2016	2019
High Tech	60.5	59.2
Medical	12.5	14.6
Other	27.0	26.2

Source: Created by case author based on data from “2019 Patent Dispute Report—Year in Review,” UnifiedPatents, January 1, 2020, accessed March 25, 2020, www.unifiedpatents.com/insights/2019/12/30/q4-2019-patent-dispute-report; “2016 Annual Patent Dispute Report,” UnifiedPatents, January 1, 2017, accessed March 25, 2020, www.unifiedpatents.com/insights/2016/12/28/2016-annual-patent-dispute-report.

EXHIBIT 4: PATENT DISPUTE VENUES BY NUMBER OF CASES (2016, 2019)

Note: NPE = non-practicing entity.

Source: Created by case author based on data from “2019 Patent Dispute Report—Year in Review,” UnifiedPatents, January 1, 2020, accessed March 25, 2020, www.unifiedpatents.com/insights/2019/12/30/q4-2019-patent-dispute-report; “2016 Annual Patent Dispute Report,” UnifiedPatents, January 1, 2017, accessed March 25, 2020, www.unifiedpatents.com/insights/2016/12/28/2016-annual-patent-dispute-report.

EXHIBIT 5: S&P 500 COMPANIES WITH THE MOST CASH (MARCH 2020)

Source: Created by case author based on data from Matt Krantz, "\$325 Billion in the Bank: Nine Companies Ready for Any Crisis," Investor's Business Daily, March 19, 2020, accessed March 25, 2020, www.investors.com/etfs-and-funds/sectors/sp500-companies-stockpile-325-billion-cash-financial-mayhem-cornavirus-stock-market-crash/.

EXHIBIT 6: COMPARISON OF PATENT GRANTS CLAIMED AND R&D SPENDING (TOP 10 COMPANIES)

Top 10 Companies with the Highest Number of Patent Grants Claimed in the United States		Top 10 Companies with the Highest R&D Spending Globally (US\$ Billions)	
Company	2019	Company	2018
IBM	9,262	Amazon	22.6
Samsung	6,469	Alphabet	16.2
Canon	3,548	Volkswagen	15.8
Microsoft	3,081	Samsung	15.3
Intel	3,020	Intel Co	13.1
LG Electronics	2,805	Microsoft	12.3
Apple	2,490	Apple	11.6
Ford Global Technologies	2,468	Roche	10.8
Amazon	2,427	Johnson & Johnson	10.6
Huawei Technologies	2,418	Merck	10.2

Source: Created by case author based on data from Erin Duffin, "Most Innovative Companies in the World, by Number of Patent Grants Claimed in the US in 2019," Statista, January 30, 2020, accessed March 25, 2020, www.statista.com/statistics/227230/worlds-most-innovative-companies/; Erin Duffin, "Ranking of the 20 Companies with the Highest Spending on Research and Development in 2018," Statista, October 2018, updated July 22, 2020, accessed March 25, 2020, www.statista.com/statistics/265645/ranking-of-the-20-companies-with-the-highest-spending-on-research-and-development/.

ENDNOTES

- ¹ This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Apple Inc. or any of its employees.
- ² Patent troll firms used patents “as legal weapons, instead of actually creating any new products or coming up with new ideas.” They frequently bought patents at discounts from failing businesses that were “looking to monetize what resources they have left, such as patents,” “Patent Trolls,” E30FF: Electronic Frontier Foundation, March 16, 2012, accessed March 17, 2020, www.eff.org/issues/resources-patent-troll-victims.
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- ⁵ Ibid.
- ⁶ Ed Hardy, “Apple, Samsung Crush a Troll’s Hotspot Patent,” Cult of Mac, September 11, 2018, accessed March 17, 2020, www.cultofmac.com/574899/apple-samsung-crush-patent-troll-hotspot/; Malcolm Owen, “Apple Urges EU to Halt the Rise of Patent Trolls,” AppleInsider, January 16, 2020, accessed March 17, 2020, <https://appleinsider.com/articles/20/01/16/apple-urges-eu-to-halt-the-rise-of-patent-trolls>.
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- ¹⁴ Ben Winck, “Apple Surpasses Microsoft as the World’s Most Valuable Company (AAPL, MSFT),” Markets Insider, October 18, 2019, accessed March 17, 2020, <https://markets.businessinsider.com/news/stocks/apple-market-cap-surpasses-microsoft-becomes-worlds-most-valuable-company-2019-10-1028611689>.
- ¹⁵ Luke Dormehl, “Apple’s Pile of Money is Patent Trolls’ Favorite Target,” Cult of Mac, July 13, 2015, accessed March 17, 2020, www.cultofmac.com/328941/apple-is-the-company-most-targeted-by-patent-trolls/.
- ¹⁶ All currency figures are in US\$ unless otherwise specified; “Global Revenue of Apple from 2004 to 2019,” Statista, October 2019, accessed March 17, 2020, www.statista.com/statistics/265125/total-net-sales-of-apple-since-2004/.
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- ¹⁸ Nils-Gerrit Wunsch, “Number of Apple Patent Families Worldwide by Filing Year from 1998 to 2018, by Legal Status,” Statista, July 2019, updated September 8, 2020, accessed March 25, 2020, www.statista.com/statistics/1033935/number-of-apple-patents-by-filing-year-and-status-worldwide/.
- ¹⁹ Dormehl, op. cit.
- ²⁰ Nick Statt, “Apple Ordered to Pay Patent Troll More than \$500 Million in iMessage Case,” The Verge, April 10, 2018, accessed March 17, 2020, www.theverge.com/2018/4/10/17222380/apple-virnetx-patent-troll-litigation-500-million-imessage-facetime-case.
- ²¹ Ibid.
- ²² VirnetX Holding Corporation, “Form 10-K: Annual Report Pursuant to Section 13 of 15(d) of the Securities Exchange Act of 1934 for the Fiscal Year Ended December 31, 2017,” United States Securities and Exchange Commission, 2018, accessed March 17, 2020, www.sec.gov/Archives/edgar/data/1082324/000114036118013886/h10051574x1_10k.htm.
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